Urban Development Assessment in Pune District Using Multi-Source Geospatial Data and Spatial Analytics

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Introduction

Rapid urbanization has significantly transformed Pune District, resulting in changes in land use, infrastructure development, vegetation cover, and population distribution. Monitoring these changes requires an integrated geospatial approach rather than relying on a single indicator. This study combines remote sensing, GIS, spatial databases, and statistical analysis to evaluate urban development using multiple environmental, demographic, and infrastructure indicators.

Aim

- To develop a comprehensive GIS-based Urban Development Assessment Framework for Pune District by integrating multi-source geospatial datasets and statistical techniques to evaluate spatial patterns of urban development and support evidence-based planning.

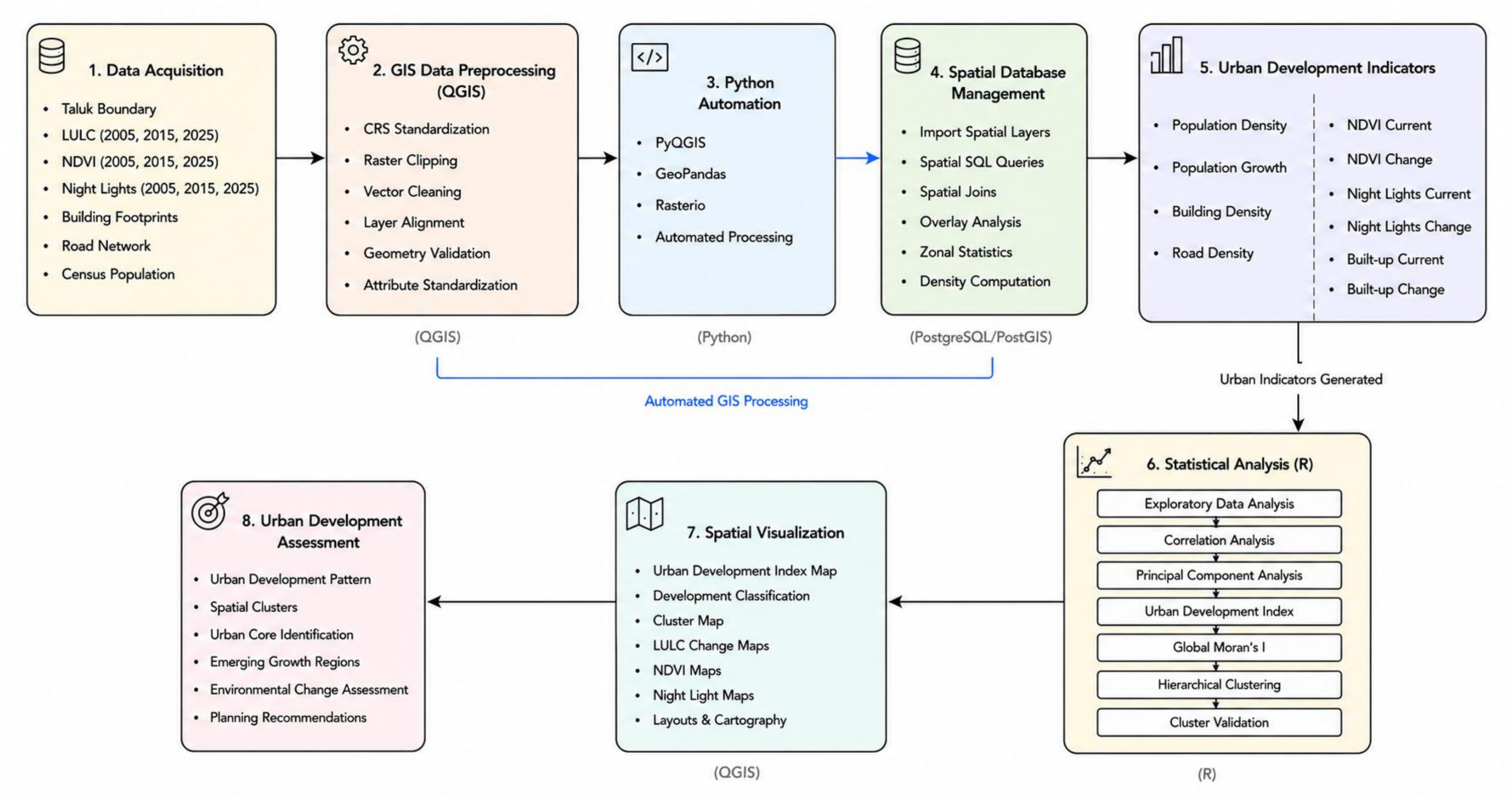
Objectives

- Quantify spatial variation in urban development across Pune's 12 taluks using multi-source geospatial indicators
- Construct a composite Urban Development Index (UDI) using PCA-based variance weighting
- Identify spatial clusters of development intensity using global Moran's I
- Classify taluks into empirically derived typologies using hierarchical clustering

Research Questions

- How does urban development intensity vary spatially across Pune District?
- Which indicators are the strongest predictors of urbanisation?
- Are spatially adjacent taluks developing at similar rates (spatial dependence)?

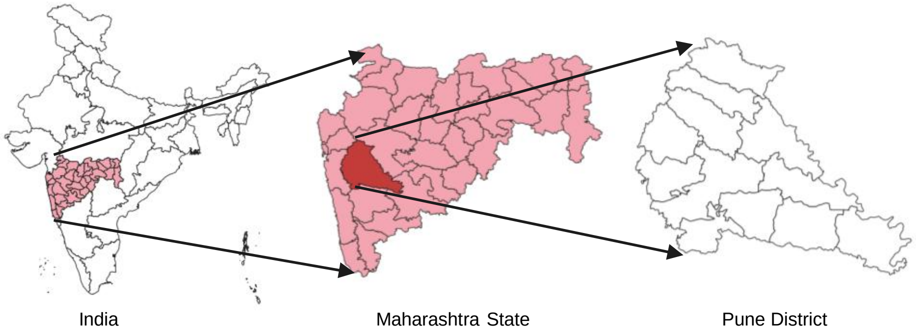
Methodology



Study Area

Pune District is one of India's fastest-growing metropolitan regions, experiencing rapid urban expansion driven by population growth, infrastructure development, and industrialization. Its diverse urban and rural landscape makes it an ideal case study for evaluating spatial patterns of urban development using integrated geospatial datasets and statistical techniques.

- **Pune District, Maharashtra**
- **Location:** Western Maharashtra, India
- **Area:** 15,643 km^2
- **Administrative Units**: 12 Taluks
- **Major Urban Centre:** Pune City
- **Projection Used:** WGS 84 / UTM Zone 43N (EPSG:32643)
- **Study Period:** 2005–2025



Pune District with Taluks

Datasets & Sources, Tools

Dataset	Year	Type	Source	Purpose
Taluk Boundary	Current	Vector	Survey of India / Administrative Boundary	Study area and zonal analysis
Land Use / Land Cover	2005, 2015, 2025	Raster	India-WRIS, ESRI Living Atlas (10 m)	Built-up mapping and land use change
NDVI	2005, 2015, 2025	Raster	Sentinel-2 / Landsat	Vegetation assessment
Night-Time Lights	2005, 2015, 2025	Raster	VIIRS Night-Time Lights	Urban activity and economic intensity
Building Footprints	Current	Vector	OpenStreetMap	Building density estimation
Road Network	Current	Vector	OpenStreetMap	Road density analysis
Population	Census 2011	Tabular	Census of India	Population density calculation
Population Growth	2001–2011	Tabular	Census of India	Demographic growth analysis

GIS & Database Tools

- **QGIS –** GIS processing and cartography
- **PostgreSQL –** Spatial database management
- **PostGIS –** Spatial SQL analysis
- **Google Earth Engine –** Satellite image processing
- **ggplot2 –** Data visualization

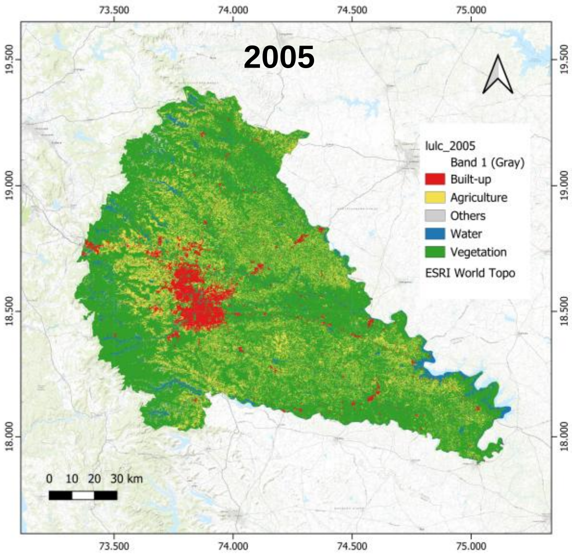
Python Libraries

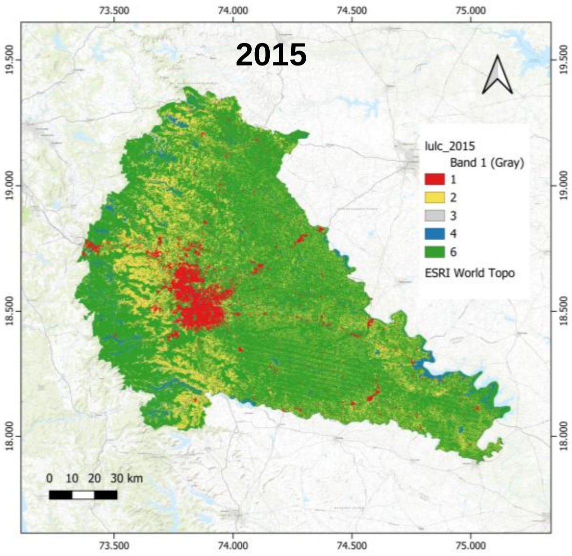
- **PyQGIS –** GIS automation
- **GeoPandas – Spatial vector analysis**
- **Rasterio –** Raster processing
- **Pandas –** Data manipulation
- **NumPy –** Numerical computation

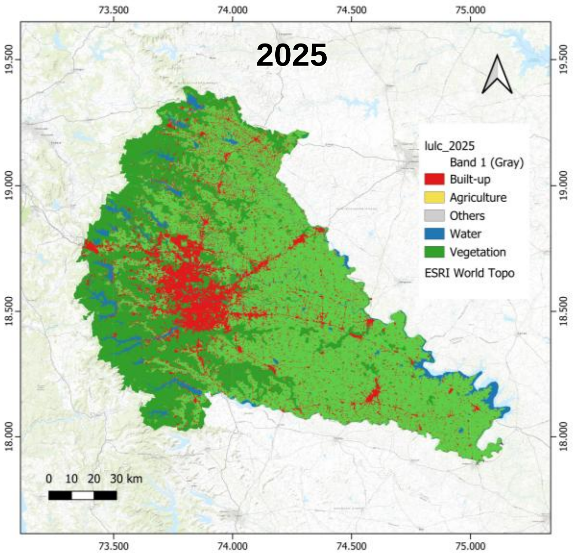
R Packages

- **sf –** Spatial data handling
- **FactoMineR / factoextra –** PCA analysis
- **spdep** – Moran's I spatial statistics
- **cluster –** Hierarchical clustering

Land Use / Land Cover Change (2005–2025)



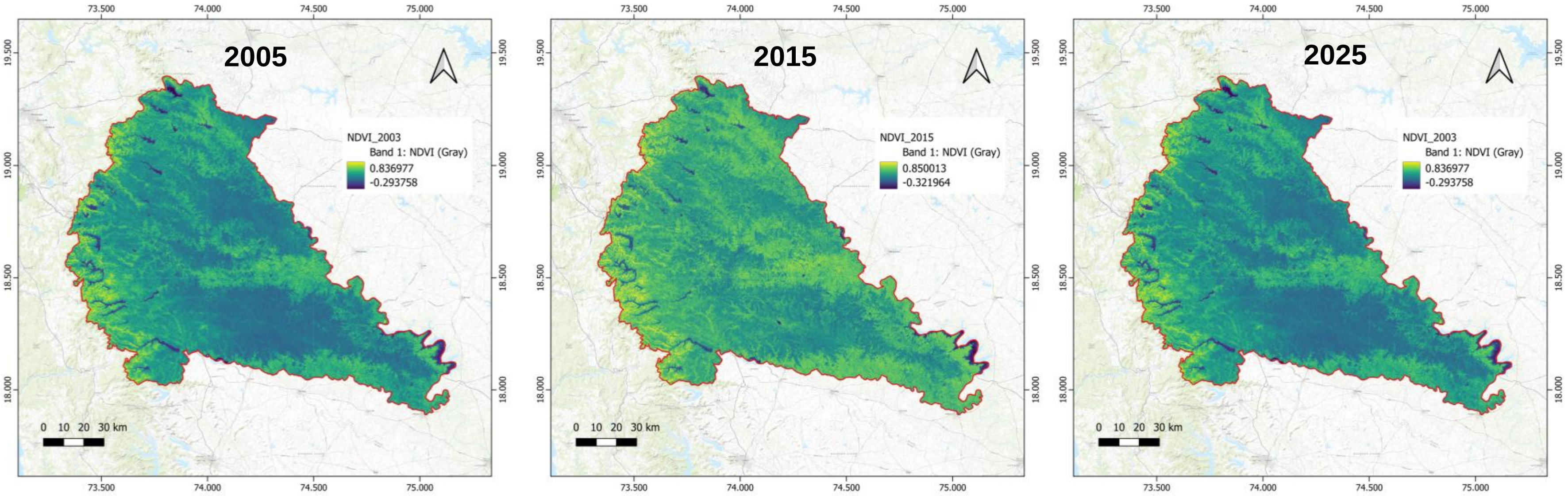




- Built-up areas expanded substantially around Pune City and along the major transportation corridors between 2005 and 2025, indicating continued urban expansion.
- Agricultural land and vegetation remain the dominant land cover across the district, although localized conversion to urban land is evident around the metropolitan region.
- Water bodies exhibit relatively limited temporal variation compared to the expansion of built-up areas.

Source: India-WRIS, ESRI Living Atlas (10 m)

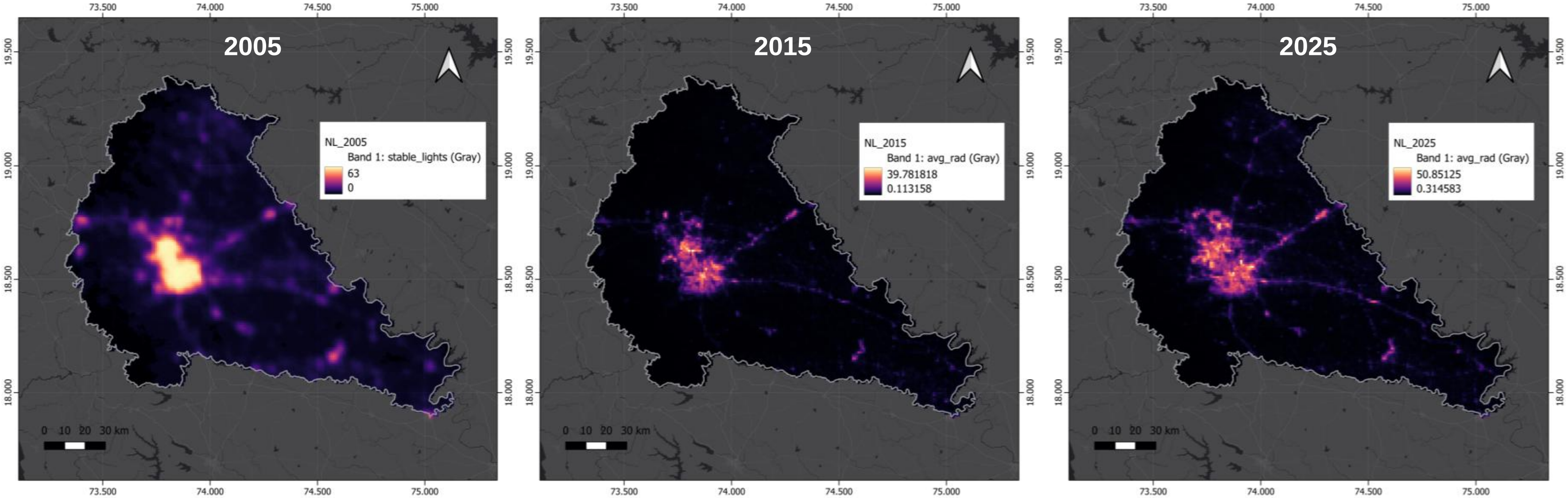
NDVINormalized Difference Vegetation Index (NDVI) (2005–2025)



- Most of Pune District maintained moderate to high vegetation cover throughout the study period.
- Lower NDVI values are concentrated around the urban core of Pune City, reflecting reduced vegetation associated with dense urban development.
- Overall vegetation patterns remain relatively stable at the district scale, with localized declines observed near expanding built-up areas.

Source: VIIRS Landsat8

Night-Time Light Intensity (2005–2025)



- High night-time light intensity is consistently concentrated around Pune City, highlighting the district’s primary urban and economic center.
- Progressive expansion of illuminated corridors indicates continued infrastructure development and urban growth toward peri-urban areas.
- Night-time lights effectively capture the spatial footprint of urbanization and complement the built-up and population indicators used in the Urban Development Index.

Source: VIIRS Night-Time Lights

Urban Development Indicators

Indicator	Method / Source	Urban Trend	Planning Significance
Population Density	Census Population / Taluk Area (km²)	⬆ Higher in urban centres	Demand for housing & public services
Population Growth	Population Change (2001–2011)	⬆ Peri-urban expansion	Identifies growth corridors
Building Density	Building Footprint Area / Taluk Area	⬆ Dense urban fabric	Urban compactness & land utilization
Road Density	Total Road Length / Taluk Area	⬆ Better accessibility	Infrastructure connectivity
NDVI (Current)	Mean NDVI (2025)	⬇ Lower in built-up areas	Vegetation and environmental quality
NDVI Change	$NDVI_{2025} - NDVI_{2005}$	⬇ Negative = vegetation loss	Environmental impact of urbanization
Night-Time Lights (Current)	Mean VIIRS Radiance (2025)	⬆ Higher economic activity	Urban intensity & commercial centres
Night-Time Lights Change	$NL_{2025} - NL_{2015}$	⬆ Expanding illuminated areas	Emerging development zones
Built-up Area (Current)	% Built-up Area (LULC 2025)	⬆ Greater urban extent	Existing urban footprint
Built-up Change	$\text{Built-up}_{2025} - \text{Built-up}_{2005}$	⬆ Urban expansion	Sprawl and land conversion

- Total Indicators: 10
- Temporal Coverage: 2005–2025
- Spatial Unit: Taluk (12 Taluks)
- Data Sources: Census of India, ESRI LULC, Google Earth Engine NDVI, VIIRS Night-Time Lights, OpenStreetMap
- Purpose: Multi-dimensional assessment of urban development using demographic, infrastructure, environmental, and urbanization indicators.

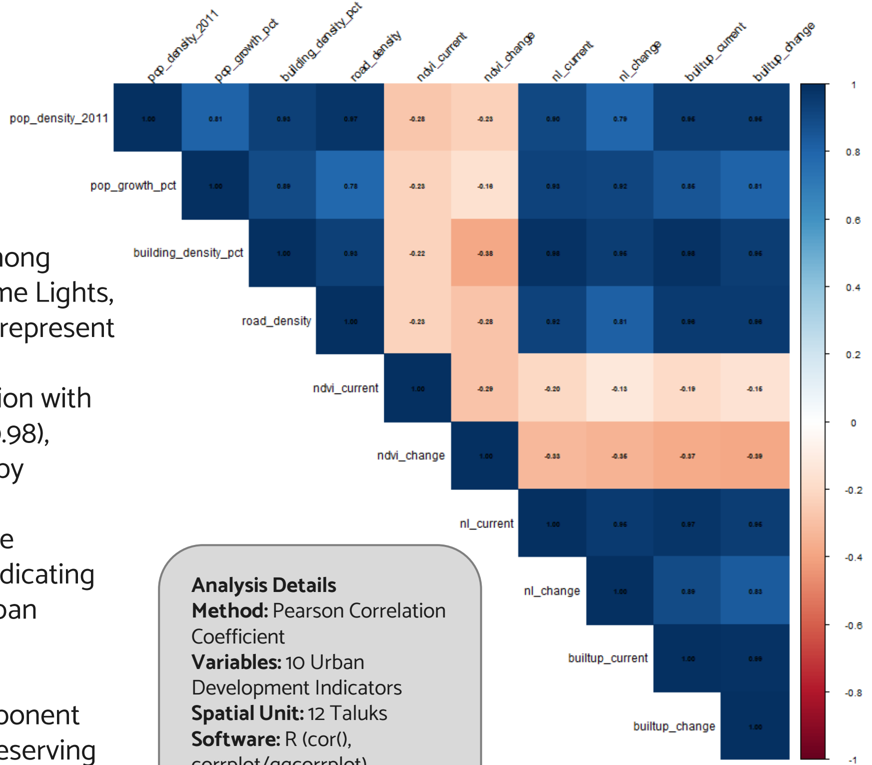
Correlation Analysis

Objective

- To evaluate the linear relationships among the selected urban development indicators and identify redundant variables prior to Principal Component Analysis (PCA).

Key Findings

- Strong positive correlations (r = 0.78–0.98) were observed among Population Density, Building Density, Road Density, Night-Time Lights, and Built-up Area, indicating that these variables collectively represent the intensity of urban development.
- Built-up Area (Current) exhibited a very high positive correlation with Night-Time Lights (Current) (r ≈ 0.97) and Road Density (r ≈ 0.98), suggesting that increasing urban expansion is accompanied by improved infrastructure and higher economic activity.
- NDVI (Current) and NDVI Change showed consistent negative correlations with urbanization indicators (r = -0.13 to -0.39), indicating that vegetation cover generally decreases with increasing urban development.
- The presence of several highly correlated variables indicates multicollinearity within the dataset. Therefore, Principal Component Analysis (PCA) was employed to reduce redundancy while preserving the dominant variance structure.



Analysis Details
Method: Pearson Correlation Coefficient
Variables: 10 Urban Development Indicators
Spatial Unit: 12 Taluks
Software: R (cor(), corrplot/ggcorrplot)

Principal Component Analysis (PCA)

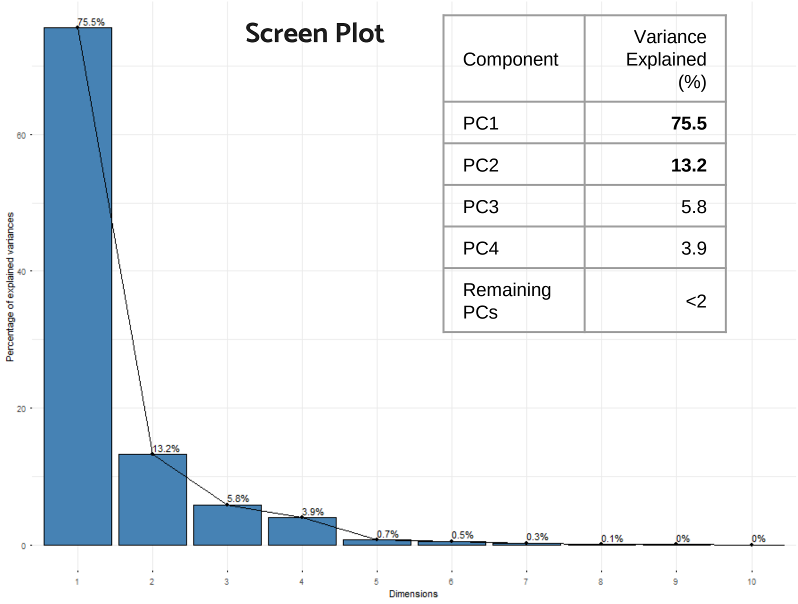
Objective
To reduce redundancy among highly correlated urban development indicators and derive a smaller set of independent principal components that capture the maximum variance in the dataset.

Interpretation

- Principal Component Analysis (PCA) was applied to the 10 standardized urban development indicators.
- The method transforms correlated variables into orthogonal principal components, preserving the maximum amount of variance.
- PC1 explained 75.5% of the total variance, indicating that a single component captures the dominant urban development pattern.
- The first two components together explained 88.7% of the dataset variance, demonstrating that most of the information is retained after dimensionality reduction.

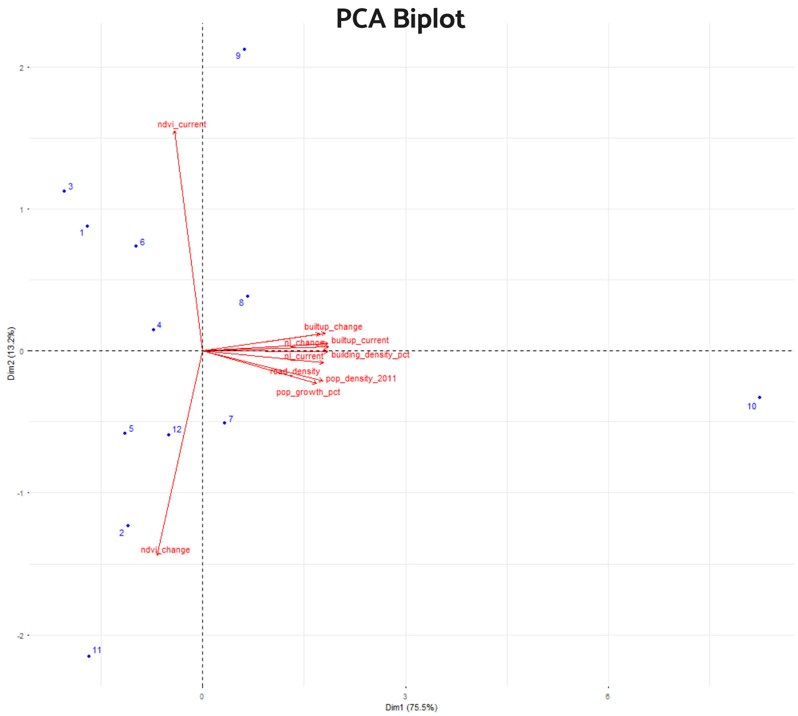
Software: R (prcomp)
Input Variables: 10
Spatial Units: 12 Taluks
Standardization: Z-score Normalization

Screen Plot



Component	Variance Explained (%)
PC1	**75.5**
PC2	**13.2**
PC3	5.8
PC4	3.9
Remaining PCs	<2

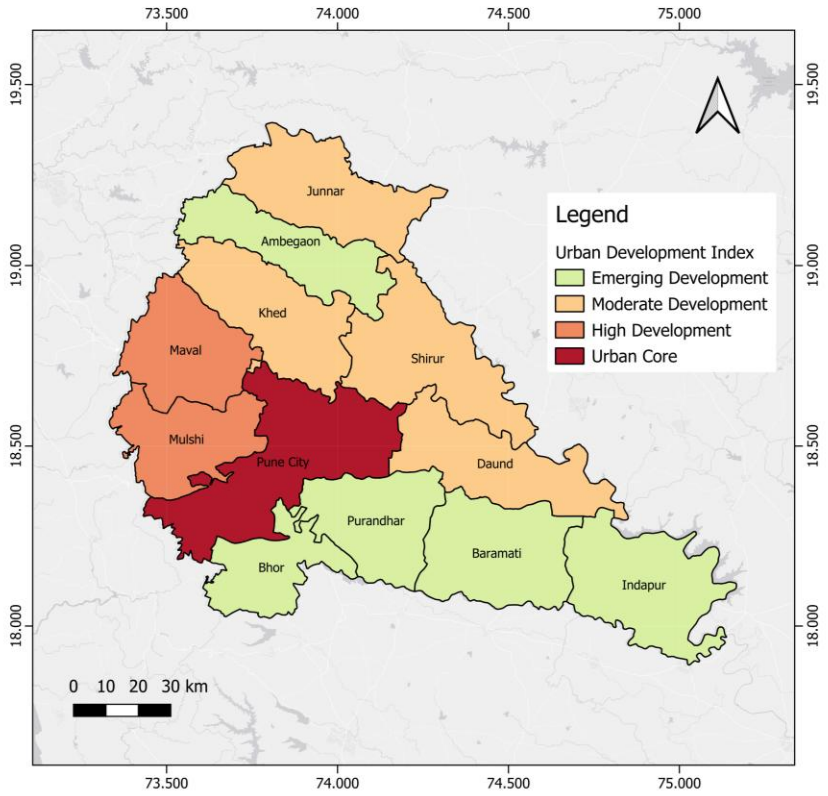
Interpretation of Principal Components



Component Interpretation

- PC1 represents the Urban Development Gradient, with strong positive loadings from Population Density, Building Density, Road Density, Built-up Area, and Night-Time Lights.
- Indicators related to urban infrastructure and built-up expansion are oriented in the same direction, indicating strong positive associations among them.
- NDVI Current and NDVI Change are oriented in the opposite direction, confirming their inverse relationship with urban development.
- Taluks positioned farther along the positive side of PC1 exhibit higher levels of urbanization, while taluks on the opposite side retain greater vegetation cover and lower development intensity.
- PC1 scores were selected as the Urban Development Index (UDI), representing the overall level of urban development across Pune District.

Composite Urban Development Assessment of Pune District using PCA



Urban Development Index Generation

- Ten standardized urban development indicators were integrated using Principal Component Analysis (PCA).
- The first principal component (PC1), explaining 75.5% of the total variance, was selected to represent the overall urban development pattern.
- PC1 scores were normalized using Min–Max Scaling to generate the Urban Development Index (0–100).

- Pune City emerged as the dominant urban center with the highest UDI (100).
- Maval and Mulshi represent rapidly developing peri-urban regions influenced by metropolitan expansion.
- Khed and Shirur exhibit moderate development, indicating transitional urban growth.
- Bhor, Ambegaon, and Purandhar recorded the lowest UDI values, representing predominantly rural development characteristics.

	Taluk	UDI
1	**Pune City**	**100**
2	Maval	26.33
3	Mulshi	25.92
4	Khed	23.03
5	Shirur	14.97

Spatial Autocorrelation Analysis (Global Moran's I)

Objective
To evaluate whether urban development indicators exhibit significant spatial clustering or dispersion across Pune District.

Moran's I Results

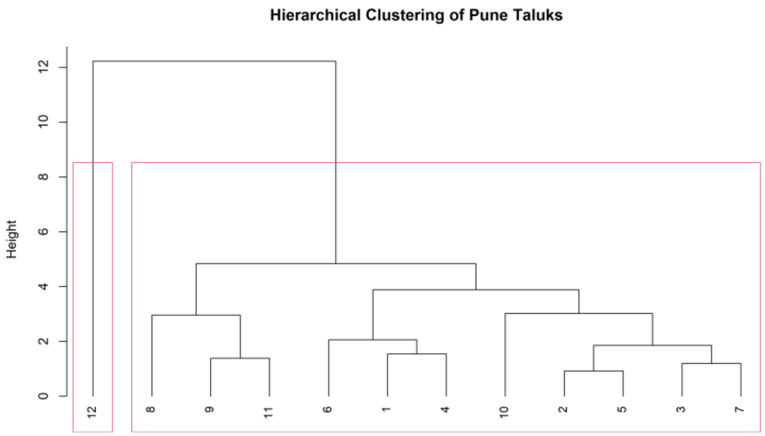
Indicator	Moran's I	P-value	Interpretation
Population Density	-0.224	0.990	Random
Building Density	0.009	0.206	Random
Road Density	-0.230	0.959	Random
NDVI Current	-0.125	0.523	Random
Night Lights	-0.007	0.236	Random
Built-up Area	-0.023	0.237	Random

- Queen Contiguity spatial weights were generated using adjacent taluk boundaries.
- Global Moran's I was computed for each urban development indicator.
- Statistical significance was evaluated using 999 Monte Carlo permutations.
- A significance level of $\alpha = 0.05$ was adopted.

Interpretation
- All indicators produced p-values greater than 0.05, indicating no statistically significant spatial autocorrelation.
- Moran's I values close to zero suggest that urban development indicators are randomly distributed at the taluk level.
- The absence of significant clustering indicates substantial spatial heterogeneity in urban development across Pune District.

Hierarchical Clustering Analysis of Pune Taluks

- Hierarchical clustering was performed on the 10 standardized urban development indicators.
- Euclidean Distance was used to measure similarity between taluks.
- Ward’s Minimum Variance (Ward.D2) linkage was applied to minimize within-cluster variance.
- The dendrogram was interpreted by retaining three major clusters.



Cluster	Taluks	Characteristics
Cluster 1	**Pune City**	Metropolitan core with the highest urban development intensity.
Cluster 2	**Mulshi, Khed, Maval**	Peri-urban growth corridor with expanding infrastructure and built-up development.
Cluster 3	**Remaining Taluks**	Predominantly rural regions with comparatively lower urbanization levels.

- Pune City forms an independent cluster due to its exceptionally high urban intensity and infrastructure concentration.
- Mulshi, Maval, and Khed exhibit similar development characteristics, reflecting the influence of metropolitan expansion.
- The remaining taluks represent a broader emerging development group with relatively lower urbanization and stronger agricultural characteristics.

Conclusion

- Integrated 10 multi-source geospatial indicators across demographic, infrastructure, and remote sensing domains into a single, variance-weighted Urban Development Index, with PC1 capturing 75.5% of total variance – demonstrating strong dimensional coherence in Pune's urbanisation signal
- Hierarchical clustering (Ward.D2) identified three empirically grounded development typologies: Urban Core (Pune City), Peri-Urban Transition (Maval, Mulshi, Khed), and Emerging/Rural Hinterland – providing a spatially differentiated evidence base beyond simple ranking
- The PostGIS–Python–R pipeline establishes a reproducible, scalable geospatial framework applicable to any Indian district with census and satellite data availability
- Non-significant Global Moran's I ($p > 0.05$) at the taluk scale indicates that development intensity does not follow a spatially structured neighbourhood pattern at this administrative resolution – a finding with direct implications for decentralised planning strategy
- The composite UDI framework offers a transferable methodology for evidence-based regional planning, smart city investment prioritisation, and SDG 11 monitoring at sub-district scale

Github : https://github.com/kondojukushi10-rgb/pune-urban-development-index

Limitations

- Taluk-level aggregation masks intra-unit spatial heterogeneity – ward or village-level analysis would reveal localised development pressure and infrastructure gaps not visible at this resolution
- Temporal comparability of night-time light data is constrained by sensor discontinuity between DMSP-OLS (2005) and VIIRS (2015, 2025), as differences in radiometric sensitivity and saturation characteristics may partly reflect instrument change rather than actual luminosity change
- The composite index excludes critical socio-economic dimensions – land value gradients, employment density, income levels, and transport accessibility – limiting its capacity to capture the full spectrum of urban development
- All demographic indicators are anchored to the 2011 Census baseline; post-2011 population dynamics, particularly accelerated peri-urban migration, are represented only through remote sensing proxies rather than direct measurement